Arunabha Majumder

Robotics Systems Engineer

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EDUCATION

PhD, Mechatronics and Robotics
Aalborg University | 2023 - present
Thesis expected March

MTech, Mechatronics
AcSIR - CSIR-CMERI, India
2020 - 2022

BTech, Mechanical Engineering
Siddaganga Institute of Technology, India
2015 - 2019

Senior Diploma in Fine Arts (Painting)
Bangiya Sangeet Parishad
2008, First Division with Distinction

STUDY ABROAD

National University of Singapore
Sep - Nov 2025 (during my PhD)

SKILLS

CAD: SolidWorks (CSWA), technical drawings, stiffness analysis in SolidWorks, Fusion 360 (basic, last used 2019)

Systems: needs to measurable targets, torque and force budgets, kinematics, Jacobian-based force estimation, model vs test verification, GA optimisation

Build and test: 3D printing, machining (Bachelor's), assembly, test setups, motors, encoders, force/torque sensors, Arduino, MATLAB/Simulink, LabVIEW

Art and design: hand sketching, painting, Adobe Illustrator, Photoshop, Procreate

ACHIEVEMENTS

Best Paper Award, IFToMM ISRM 2026
Best Major Project Award, 2019
TATA Technologies Prayatna grant
GATE qualified

LANGUAGES

English (C1), Bengali (native), Hindi

REFERENCES

Prof. Shaoping Bai
Aalborg University, sbh@mp.aau.dk

Assoc. Prof. Lars Rosgaard Jensen
Head of Section and Head of Labs, AAU

Assoc. Prof. Anderson de Souza Castelo Oliveira
Materials and Production, AAU

PROFILE

Robotics engineer who turns human needs into measurable targets, integrates mechanics, actuation, sensing and control, and proves performance on real hardware. Award-winning shoulder exoskeleton: torque shared between a spring mechanism and a 6 Nm motor, 25%+ less motor energy. 5-bar robot stiffness model verified within 10% of tests. 6 years of medical and assistive robotics with human-subject testing, working across mechanical, electrical, software and controls.

EXPERIENCE

Research Assistant
Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid shoulder exoskeleton: a variable stiffness mechanism (spring pretensioned by a worm gear) working in parallel with a 6 Nm motor; together they give the torque needed to lift the arm. It uses over 25% less motor energy (Best Paper Award, IFToMM ISRM 2026).
- Optimised the mechanism with a genetic algorithm (diameter, spring stiffness, pretension force), made the CAD in SolidWorks, 3D printed prototypes, had final parts manufactured.
- Integrated the motor, encoder and force/torque sensors (actuation and sensing) and verified the stiffness the system can provide on a physical test setup.
- Designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM); estimated its stiffness with less than 10% error and quantified human arm impedance. Force estimation used the variable stiffness actuators and the robot's Jacobian.
- Arm impedance study with the 5-bar robot (6 participants): Qualisys cameras tracked the arm, and I implemented an Arduino-based trigger to sync the robot's motion with the recording.
- Taught courses and mentored Bachelor's students (2023 - 2025).

MTech Researcher, Soft Exosuit
CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic artificial muscles for a soft exosuit. Designed the outer PID loop, with integral anti-windup and error nonlinearisation, of a neural network-based gain-scheduled controller in MATLAB/Simulink and implemented it in LabVIEW to drive the muscle (0.3 - 0.78% error).

Bachelor's Projects
Siddaganga Institute of Technology, India | *2017 - 2019*

- Built an arm exoskeleton and exo glove for a paralysed arm: CAD, machining, assembly and testing.
- Built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018) with student teams.

PUBLICATIONS

- Hybrid actuated shoulder exoskeleton, IFToMM ISRM 2026 (Springer), Best Paper Award.
- Neural network-based control of a pneumatic artificial muscle, IEEE CONECCT 2022.

PORTFOLIO

Videos and results from my work: arunabha22.github.io/arunabha-majumder